

Week 8: Database Relationship Analysis using Joins

Scenario:

The e-commerce company stores customer, product, order, and payment information in separate tables. The management team wants to combine these related tables to understand customer purchases, order details, and payment status.

Implement INNER JOIN

Concept:

INNER JOIN returns only matching records from both tables.

Tasks:

1. Display customer details along with their orders.

```
SQL> SELECT CONCAT(C.First_Name,C.Last_Name) AS Customer_Name,
2 Orders.OrderID,
3 Orders.TotalAmount
4 FROM Customer C
5 INNER JOIN Orders
6 ON C.Customer_ID = Orders.CustomerID;
```

CUSTOMER_NAME	ORDERID	TOTALAMOUNT
ArjunKumar	ORD0001	55850
PriyaSharma	ORD0002	3350
RahulMenon	ORD0003	28000
AnanyaIyer	ORD0004	8300
VikramReddy	ORD0005	22000
MeeraKrishnan	ORD0006	7000
SureshNair	ORD0007	6500
KavyaRajan	ORD0008	5700
KarthikSubramanian	ORD0009	18500
DivyaPillai	ORD0010	5200
AdityaVerma	ORD0011	7000
SnehaPatel	ORD0012	2450
LakshmiVenkatesan	ORD0014	28000
NithinMurugan	ORD0015	8350

14 rows selected.

2. Display order details with payment information.

```

SQL> SELECT o.orderID,
2  o.TotalAmount,
3  o.OrderDate,
4  o.CustomerID,
5  p.PaymentDate,
6  p.PaymentMode,
7  p.PaymentStatus
8  FROM Payment p JOIN Orders o
9  ON p.OrderID = o.OrderID;

```

ORDERID	TOTALAMOUNT	ORDERDATE	CUSTOMERID	PAYMENTDA	PAYMENTMODE	PAYMENTSTA
ORD0001	55850	10-JAN-26	CUST000001	10-JAN-26	Credit Card	Completed
ORD0002	3350	25-JAN-26	CUST000002	25-JAN-26	UPI	Completed
ORD0004	8300	18-FEB-26	CUST000004	18-FEB-26	Debit Card	Completed
ORD0006	7000	15-MAR-26	CUST000006	15-MAR-26	Net Banking	Completed
ORD0008	5700	20-APR-26	CUST000008	20-APR-26	COD	Completed
ORD0010	5200	18-MAY-26	CUST000010	18-MAY-26	UPI	Completed
ORD0012	2450	20-JUN-26	CUST000012	20-JUN-26	Credit Card	Completed
ORD0014	28000	15-AUG-26	CUST000014	15-AUG-26	Debit Card	Completed
ORD0014	28000	15-AUG-26	CUST000014	14-AUG-26	Debit Card	Failed

9 rows selected.

- Retrieve products purchased by customers.

9 rows selected.

```
SQL> SELECT
  2     C.CUSTOMER_ID,
  3     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4     O.ORDERID AS ORDER_ID,
  5     P.PRODUCT_ID,
  6     P.PRODUCT_NAME,
  7     OD.QUANTITY,
  8     OD.PRICE,
  9     OD.QUANTITY * OD.PRICE AS TOTAL_AMOUNT
10 FROM CUSTOMER C
11 JOIN ORDERS O
12     ON C.CUSTOMER_ID = O.CUSTOMERID
13 JOIN ORDERDETAIL OD
14     ON O.ORDERID = OD.ORDERID
15 JOIN PRODUCT P
16     ON OD.PRODUCTID = P.PRODUCT_ID
17 ORDER BY C.CUSTOMER_ID, O.ORDERDATE;
```

CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME		QUANTITY	PRICE
TOTAL_AMOUNT			
CUST000001 Arjun Kumar		ORD0001	P000001
Laptop		1	55000
55000			
CUST000001 Arjun Kumar		ORD0001	P000002
Wireless Mouse		1	850
850			
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME		QUANTITY	PRICE
TOTAL_AMOUNT			
CUST000002 Priya Sharma		ORD0002	P000006
USB Cable		1	450
450			
CUST000002 Priya Sharma		ORD0002	P000004
Bluetooth Speaker		1	3200
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME		QUANTITY	PRICE
TOTAL_AMOUNT			
3200			
CUST000003 Rahul Menon		ORD0003	P000005
Smartphone		6	28000
168000			
CUST000004 Ananya Iyer		ORD0004	P000006

PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
USB Cable 1800	4	450
CUST000004 Ananya Iyer Smart Watch 6500	ORD0004 P000008 1	6500
CUSTOMER_I CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
CUST000005 Vikram Reddy Tablet 22000	ORD0005 P000010 1	22000
CUST000006 Meera Krishnan Wireless Earbuds 3500	ORD0006 P000015 1	3500
CUSTOMER_I CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
CUST000006 Meera Krishnan Headphones 4200	ORD0006 P000009 1	4200
CUST000007 Suresh Nair Smart Watch 6500	ORD0007 P000008 1	6500
CUSTOMER_I CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
6500		
CUST000008 Kavya Rajan External Hard Disk 5200	ORD0008 P000012 1	5200
CUST000008 Kavya Rajan	ORD0008 P000006	
CUSTOMER_I CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
USB Cable 450	1	450
CUST000009 Karthik Subramanian Gaming Monitor 18500	ORD0009 P000013 1	18500
CUSTOMER_I CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE
TOTAL_AMOUNT		
CUST000010 Divya Pillai External Hard Disk 5200	ORD0010 P000012 1	5200
CUST000011 Aditya Verma Wireless Earbuds 3500	ORD0011 P000015 1	3500

CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE	
TOTAL_AMOUNT			
CUST000011	Aditya Verma	ORD0011	P000009
Headphones	1	4200	
4200			
CUST000012	Sneha Patel	ORD0012	P000014
Laptop Stand	1	1500	
1500			
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE	
TOTAL_AMOUNT			
1500			
CUST000012	Sneha Patel	ORD0012	P000003
Mechanical Keyboard	1	2500	
2500			
CUST000015	Nithin Murugan	ORD0015	P000015
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	PRODUCT
PRODUCT_NAME	QUANTITY	PRICE	
TOTAL_AMOUNT			
Wireless Earbuds	1	3500	
3500			

20 rows selected.

Implement LEFT JOIN

Tasks:

1. Display all customers including customers who have not placed orders.

```

SQL> SELECT
2      C.CUSTOMER_ID,
3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4      C.EMAIL,
5      O.ORDERID AS ORDER_ID,
6      O.ORDERDATE AS ORDER_DATE,
7      O.TOTALAMOUNT AS TOTAL_AMOUNT,
8      O.ORDERSTATUS AS ORDER_STATUS
9  FROM CUSTOMER C
10 LEFT JOIN ORDERS O
11      ON C.CUSTOMER_ID = O.CUSTOMERID
12 ORDER BY C.CUSTOMER_ID;

```

CUSTOMER_I CUSTOMER_NAME

EMAIL	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
CUST000001 Arjun Kumar arjun.kumar@gmail.com	ORD0001	10-JAN-26	55850	Delivered
CUST000002 Priya Sharma priya.sharma@gmail.com	ORD0002	25-JAN-26	3350	Delivered
CUST000003 Rahul Menon rahul.menon@gmail.com	ORD0003	05-FEB-26	28000	Delivered

CUSTOMER_I CUSTOMER_NAME

EMAIL	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
CUST000004 Ananya Iyer ananya.iyer@gmail.com	ORD0004	18-FEB-26	8300	Delivered
CUST000005 Vikram Reddy vikram.reddy@gmail.com	ORD0005	02-MAR-26	22000	Processing
CUST000006 Meera Krishnan meera.krishnan@gmail.com	ORD0006	15-MAR-26	7000	Delivered

CUSTOMER_I CUSTOMER_NAME

EMAIL	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
CUST000007 Suresh Nair suresh.nair@gmail.com	ORD0007	01-APR-26	6500	Shipped
CUST000008 Kavya Rajan kavya.rajan@gmail.com	ORD0008	20-APR-26	5700	Delivered
CUST000009 Karthik Subramanian karthik.subramanian@gmail.com	ORD0009	05-MAY-26	18500	Processing

CUSTOMER_I CUSTOMER_NAME

EMAIL	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
CUST000010 Divya Pillai divya.pillai@gmail.com	ORD0010	18-MAY-26	5200	Delivered
CUST000011 Aditya Verma aditya.verma@gmail.com	ORD0011	03-JUN-26	7000	Shipped
CUST000012 Sneha Patel sneha.patel@gmail.com	ORD0012	20-JUN-26	2450	Delivered

CUSTOMER_I CUSTOMER_NAME

EMAIL	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
CUST000013 Rohan Bose rohan.bose@gmail.com				
CUST000014 Lakshmi Venkatesan lakshmi.venkatesan@gmail.com	ORD0014	15-AUG-26	28000	Delivered
CUST000015 Nithin Murugan nithin.murugan@gmail.com	ORD0015	25-SEP-26	8350	Processing

15 rows selected.

2. Find customers without any purchases.

```
SQL> SELECT
  2   C.CUSTOMER_ID,
  3   C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4   C.EMAIL,
  5   C.PHONE
  6 FROM CUSTOMER C
  7 LEFT JOIN ORDERS O
  8   ON C.CUSTOMER_ID = O.CUSTOMERID
  9 WHERE O.ORDERID IS NULL
 10 ORDER BY C.CUSTOMER_ID;
```

CUSTOMER_I	CUSTOMER_NAME
CUST000013	Rohan Bose
rohan.bose@gmail.com	9654321098

3. Display all products including products with no sales.

```
SQL> SELECT
  2   P.PRODUCT_ID,
  3   P.PRODUCT_NAME,
  4   P.PRICE
  5 FROM PRODUCT P
  6 LEFT JOIN ORDERDETAIL OD
  7   ON P.PRODUCT_ID = OD.PRODUCTID
  8 WHERE OD.PRODUCTID IS NULL
  9 ORDER BY P.PRODUCT_ID;
```

PRODUCT	PRODUCT_NAME	PRICE
P000007	Power Bank	1800
P000011	Web Camera	2800

Implement RIGHT JOIN

Concept:

RIGHT JOIN returns all records from the right table and matching records from the left table.

Tasks:

1. Display all orders with customer information.

```

SQL> SELECT
2     O.ORDERID AS ORDER_ID,
3     O.ORDERDATE AS ORDER_DATE,
4     O.TOTALAMOUNT AS TOTAL_AMOUNT,
5     O.ORDERSTATUS AS ORDER_STATUS,
6     C.CUSTOMER_ID,
7     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
8     C.EMAIL
9 FROM CUSTOMER C
10 RIGHT JOIN ORDERS O
11     ON C.CUSTOMER_ID = O.CUSTOMERID
12 ORDER BY O.ORDERDATE;

```

ORDER_I	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS	CUSTOMER_I
ORD0001	10-JAN-26	55850	Delivered	CUST000001
Arjun Kumar				
arjun.kumar@gmail.com				
ORD0002	25-JAN-26	3350	Delivered	CUST000002
Priya Sharma				
priya.sharma@gmail.com				
ORD0003	05-FEB-26	28000	Delivered	CUST000003
Rahul Menon				
rahul.menon@gmail.com				
ORD0004	18-FEB-26	8300	Delivered	CUST000004
Ananya Iyer				
ananya.iyer@gmail.com				
ORD0005	02-MAR-26	22000	Processing	CUST000005
Vikram Reddy				
vikram.reddy@gmail.com				
ORD0006	15-MAR-26	7000	Delivered	CUST000006
Meera Krishnan				
meera.krishnan@gmail.com				
ORD0007	01-APR-26	6500	Shipped	CUST000007
Suresh Nair				
suresh.nair@gmail.com				
ORD0008	20-APR-26	5700	Delivered	CUST000008
Kavya Rajan				
kavya.rajan@gmail.com				
ORD0009	05-MAY-26	18500	Processing	CUST000009
Karthik Subramanian				
karthik.subramanian@gmail.com				
ORD0010	18-MAY-26	5200	Delivered	CUST000010
Divya Pillai				
divya.pillai@gmail.com				
ORD0011	03-JUN-26	7000	Shipped	CUST000011
Aditya Verma				
aditya.verma@gmail.com				
ORD0012	20-JUN-26	2450	Delivered	CUST000012
Sneha Patel				
sneha.patel@gmail.com				
ORD0014	15-AUG-26	28000	Delivered	CUST000014
Lakshmi Venkatesan				
lakshmi.venkatesan@gmail.com				
ORD0015	25-SEP-26	8350	Processing	CUST000015
Nithin Murugan				
nithin.murugan@gmail.com				

14 rows selected.

- Find orders where customer details are missing.

```

SQL> SELECT
2     O.ORDERID AS ORDER_ID,
3     O.ORDERDATE AS ORDER_DATE,
4     O.TOTALAMOUNT AS TOTAL_AMOUNT,
5     O.ORDERSTATUS AS ORDER_STATUS
6 FROM CUSTOMER C
7 RIGHT JOIN ORDERS O
8     ON C.CUSTOMER_ID = O.CUSTOMERID
9 WHERE C.CUSTOMER_ID IS NULL;

```

no rows selected

3. Display all payment records with order details.

```
SQL> SELECT
  2     P.PAYMENT_ID,
  3     P.PAYMENTDATE AS PAYMENT_DATE,
  4     P.PAYMENTMODE AS PAYMENT_MODE,
  5     P.PAYMENTSTATUS AS PAYMENT_STATUS,
  6     P.TRANSACTIONAMOUNT AS TRANSACTION_AMOUNT,
  7     O.ORDERID AS ORDER_ID,
  8     O.ORDERDATE AS ORDER_DATE,
  9     O.TOTALAMOUNT AS ORDER_AMOUNT,
 10     O.ORDERSTATUS AS ORDER_STATUS
 11 FROM ORDERS O
 12 RIGHT JOIN PAYMENT P
 13     ON O.ORDERID = P.ORDERID
 14 ORDER BY P.PAYMENTDATE;
```

PAYMENT	PAYMENT_D	PAYMENT_MODE	PAYMENT_ST	TRANSACTION_AMOUNT	ORDER_I
PAY0001	10-JAN-26	Credit Card	Completed	55850	ORD0001
10-JAN-26		55850 Delivered			
PAY0002	25-JAN-26	UPI	Completed	3350	ORD0002
25-JAN-26		3350 Delivered			
PAY0003	18-FEB-26	Debit Card	Completed	8300	ORD0004
18-FEB-26		8300 Delivered			

PAYMENT	PAYMENT_D	PAYMENT_MODE	PAYMENT_ST	TRANSACTION_AMOUNT	ORDER_I
PAY0004	15-MAR-26	Net Banking	Completed	7000	ORD0006
15-MAR-26		7000 Delivered			
PAY0005	20-APR-26	COD	Completed	5700	ORD0008
20-APR-26		5700 Delivered			
PAY0006	18-MAY-26	UPI	Completed	5200	ORD0010
18-MAY-26		5200 Delivered			

PAYMENT	PAYMENT_D	PAYMENT_MODE	PAYMENT_ST	TRANSACTION_AMOUNT	ORDER_I
PAY0007	20-JUN-26	Credit Card	Completed	2450	ORD0012
20-JUN-26		2450 Delivered			
PAY0009	14-AUG-26	Debit Card	Failed	28000	ORD0014
15-AUG-26		28000 Delivered			
PAY0008	15-AUG-26	Debit Card	Completed	28000	ORD0014
15-AUG-26		28000 Delivered			

9 rows selected.

5. Retrieve Complete Order Details

Task:

Generate a complete order report by combining:

- Customer Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status

```

SQL> SELECT
2     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
3     P.PRODUCT_NAME,
4     OD.QUANTITY,
5     O.ORDERDATE AS ORDER_DATE,
6     O.TOTALAMOUNT AS TOTAL_AMOUNT,
7     PM.PAYMENTSTATUS AS PAYMENT_STATUS
8 FROM CUSTOMER C
9 JOIN ORDERS O
10      ON C.CUSTOMER_ID = O.CUSTOMERID
11 JOIN ORDERDETAIL OD
12      ON O.ORDERID = OD.ORDERID
13 JOIN PRODUCT P
14      ON OD.PRODUCTID = P.PRODUCT_ID
15 LEFT JOIN PAYMENT PM
16      ON O.ORDERID = PM.ORDERID
17 ORDER BY O.ORDERDATE, C.CUSTOMER_ID;

```

CUSTOMER_NAME	PRODUCT_NAME	QUANTITY	ORDER_DATE	TOTAL_AMOUNT	PAYMENT_STATUS
Arjun Kumar	Laptop	1	10-JAN-26	55850	Completed
Arjun Kumar	Wireless Mouse	1	10-JAN-26	55850	Completed
Priya Sharma	USB Cable	1	25-JAN-26	3350	Completed
Priya Sharma	Bluetooth Speaker	1	25-JAN-26		
Rahul Menon	Smartphone	6	05-FEB-26	28000	
Ananya Iyer	USB Cable	4	18-FEB-26	8300	Completed
Ananya Iyer	Smart Watch	1	18-FEB-26	8300	Completed

CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
Vikram Reddy Tablet		1	02-MAR-26
22000			
Meera Krishnan Wireless Earbuds		1	15-MAR-26
7000 Completed			
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
Meera Krishnan Headphones		1	15-MAR-26
7000 Completed			
Suresh Nair Smart Watch		1	01-APR-26
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
6500			
Kavya Rajan External Hard Disk		1	20-APR-26
5700 Completed			
Kavya Rajan			
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
USB Cable		1	20-APR-26
5700 Completed			
Karthik Subramanian Gaming Monitor		1	05-MAY-26
18500			
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
Divya Pillai External Hard Disk		1	18-MAY-26
5200 Completed			
Aditya Verma Wireless Earbuds		1	03-JUN-26
7000			
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
Aditya Verma Headphones		1	03-JUN-26
7000			
Sneha Patel Laptop Stand		1	20-JUN-26

CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
2450 Completed			
Sneha Patel Mechanical Keyboard		1	20-JUN-26
2450 Completed			
Nithin Murugan			
CUSTOMER_NAME			
PRODUCT_NAME		QUANTITY	ORDER_DAT
TOTAL_AMOUNT		PAYMENT_ST	
Wireless Earbuds		1	25-SEP-26
8350			

20 rows selected.

6. Display Customer Purchase History

Tasks:

Write queries to find:

1. All products purchased by a customer.

```
SQL> SELECT
  2     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3     O.ORDERID AS ORDER_ID,
  4     O.ORDERDATE AS ORDER_DATE,
  5     P.PRODUCT_NAME,
  6     OD.QUANTITY,
  7     OD.PRICE
  8 FROM CUSTOMER C
  9 JOIN ORDERS O
10     ON C.CUSTOMER_ID = O.CUSTOMERID
11 JOIN ORDERDETAIL OD
12     ON O.ORDERID = OD.ORDERID
13 JOIN PRODUCT P
14     ON OD.PRODUCTID = P.PRODUCT_ID
15 WHERE C.CUSTOMER_ID = 'CUST000006'
16 ORDER BY O.ORDERDATE;
```

CUSTOMER_NAME	ORDER_I	ORDER_DAT		
PRODUCT_NAME			QUANTITY	PRICE
Meera Krishnan	ORD0006	15-MAR-26		
Wireless Earbuds			1	3500
Meera Krishnan	ORD0006	15-MAR-26		
Headphones			1	4200

2. Total amount spent by each customer.

```
SQL> SELECT
  2     C.CUSTOMER_ID,
  3     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4     ROUND(SUM(O.TOTALAMOUNT), 2) AS TOTAL_AMOUNT_SPENT
  5 FROM CUSTOMER C
  6 JOIN ORDERS O
  7     ON C.CUSTOMER_ID = O.CUSTOMERID
  8 GROUP BY
  9     C.CUSTOMER_ID,
10     C.FIRST_NAME,
11     C.LAST_NAME
12 ORDER BY TOTAL_AMOUNT_SPENT DESC;
```

CUSTOMER_I	CUSTOMER_NAME	TOTAL_AMOUNT_SPENT
CUST000001	Arjun Kumar	55850
CUST000014	Lakshmi Venkatesan	28000
CUST000003	Rahul Menon	28000
CUST000005	Vikram Reddy	22000
CUST000009	Karthik Subramanian	18500
CUST000015	Nithin Murugan	8350
CUST000004	Ananya Iyer	8300
CUST000006	Meera Krishnan	7000
CUST000011	Aditya Verma	7000
CUST000007	Suresh Nair	6500
CUST000008	Kavya Rajan	5700
CUSTOMER_I	CUSTOMER_NAME	TOTAL_AMOUNT_SPENT
CUST000010	Divya Pillai	5200
CUST000002	Priya Sharma	3350
CUST000012	Sneha Patel	2450

14 rows selected.

3. Number of orders placed by each customer.

```
SQL> SELECT
  2   C.CUSTOMER_ID,
  3   C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4   COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
  5 FROM CUSTOMER C
  6 LEFT JOIN ORDERS O
  7   ON C.CUSTOMER_ID = O.CUSTOMERID
  8 GROUP BY
  9   C.CUSTOMER_ID,
10   C.FIRST_NAME,
11   C.LAST_NAME
12 ORDER BY NUMBER_OF_ORDERS DESC;
```

CUSTOMER_I	CUSTOMER_NAME	NUMBER_OF_ORDERS
CUST000001	Arjun Kumar	1
CUST000015	Nithin Murugan	1
CUST000003	Rahul Menon	1
CUST000004	Ananya Iyer	1
CUST000005	Vikram Reddy	1
CUST000006	Meera Krishnan	1
CUST000007	Suresh Nair	1
CUST000008	Kavya Rajan	1
CUST000009	Karthik Subramanian	1
CUST000010	Divya Pillai	1
CUST000011	Aditya Verma	1
CUSTOMER_I	CUSTOMER_NAME	NUMBER_OF_ORDERS
CUST000012	Sneha Patel	1
CUST000002	Priya Sharma	1
CUST000014	Lakshmi Venkatesan	1
CUST000013	Rohan Bose	0

15 rows selected.

4. Latest purchase details of customers.

```

SQL> SELECT
  2     C.CUSTOMER_ID,
  3     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4     O.ORDERID AS ORDER_ID,
  5     O.ORDERDATE AS LATEST_PURCHASE_DATE,
  6     O.TOTALAMOUNT AS TOTAL_AMOUNT,
  7     O.ORDERSTATUS AS ORDER_STATUS
  8 FROM CUSTOMER C
  9 JOIN ORDERS O
10     ON C.CUSTOMER_ID = O.CUSTOMERID
11 WHERE O.ORDERDATE = (
12     SELECT MAX(O2.ORDERDATE)
13     FROM ORDERS O2
14     WHERE O2.CUSTOMERID = C.CUSTOMER_ID
15 )
16 ORDER BY C.CUSTOMER_ID;

```

CUSTOMER_I	CUSTOMER_NAME	ORDER_I	LATEST_PU

TOTAL_AMOUNT	ORDER_STATUS		

CUST000001	Arjun Kumar 55850 Delivered	ORD0001	10-JAN-26
CUST000002	Priya Sharma 3350 Delivered	ORD0002	25-JAN-26
CUST000003	Rahul Menon 28000 Delivered	ORD0003	05-FEB-26
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	LATEST_PU

TOTAL_AMOUNT	ORDER_STATUS		

CUST000004	Ananya Iyer 8300 Delivered	ORD0004	18-FEB-26
CUST000005	Vikram Reddy 22000 Processing	ORD0005	02-MAR-26
CUST000006	Meera Krishnan 7000 Delivered	ORD0006	15-MAR-26
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	LATEST_PU

TOTAL_AMOUNT	ORDER_STATUS		

CUST000007	Suresh Nair 6500 Shipped	ORD0007	01-APR-26
CUST000008	Kavya Rajan 5700 Delivered	ORD0008	20-APR-26
CUST000009	Karthik Subramanian 18500 Processing	ORD0009	05-MAY-26
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	LATEST_PU

TOTAL_AMOUNT	ORDER_STATUS		

CUST000010	Divya Pillai 5200 Delivered	ORD0010	18-MAY-26
CUST000011	Aditya Verma 7000 Shipped	ORD0011	03-JUN-26
CUST000012	Sneha Patel 2450 Delivered	ORD0012	20-JUN-26
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	LATEST_PU

TOTAL_AMOUNT	ORDER_STATUS		

CUST000014	Lakshmi Venkatesan 28000 Delivered	ORD0014	15-AUG-26
CUST000015	Nithin Murugan 8350 Processing	ORD0015	25-SEP-26

14 rows selected.

7. Generate Multi-Table Business Reports

Report 1: Customer Order Report

Display:

- Customer name
- Order ID
- Order date
- Order status

```
SQL> SELECT
  2     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3     O.ORDERID AS ORDER_ID,
  4     O.ORDERDATE AS ORDER_DATE,
  5     O.ORDERSTATUS AS ORDER_STATUS
  6 FROM CUSTOMER C
  7 JOIN ORDERS O
  8     ON C.CUSTOMER_ID = O.CUSTOMERID
  9 ORDER BY O.ORDERDATE;
```

CUSTOMER_NAME	ORDER_I	ORDER_DAT	ORDER_STATUS
Arjun Kumar	ORD0001	10-JAN-26	Delivered
Priya Sharma	ORD0002	25-JAN-26	Delivered
Rahul Menon	ORD0003	05-FEB-26	Delivered
Ananya Iyer	ORD0004	18-FEB-26	Delivered
Vikram Reddy	ORD0005	02-MAR-26	Processing
Meera Krishnan	ORD0006	15-MAR-26	Delivered
Suresh Nair	ORD0007	01-APR-26	Shipped
Kavya Rajan	ORD0008	20-APR-26	Delivered
Karthik Subramanian	ORD0009	05-MAY-26	Processing
Divya Pillai	ORD0010	18-MAY-26	Delivered
Aditya Verma	ORD0011	03-JUN-26	Shipped

CUSTOMER_NAME	ORDER_I	ORDER_DAT	ORDER_STATUS
Sneha Patel	ORD0012	20-JUN-26	Delivered
Lakshmi Venkatesan	ORD0014	15-AUG-26	Delivered
Nithin Murugan	ORD0015	25-SEP-26	Processing

14 rows selected.

Report 2: Sales Report

Display:

- Product name
- Quantity sold
- Total revenue

```
SQL> SELECT
  2     P.PRODUCT_NAME,
  3     SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  4     SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  5 FROM PRODUCT P
  6 JOIN ORDERDETAIL OD
  7     ON P.PRODUCT_ID = OD.PRODUCTID
  8 GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9 ORDER BY TOTAL_REVENUE DESC;
```

PRODUCT_NAME	QUANTITY_SOLD	TOTAL_REVENUE
Smartphone	6	168000
Laptop	1	55000
Tablet	1	22000
Gaming Monitor	1	18500
Smart Watch	2	13000
Wireless Earbuds	3	10500
External Hard Disk	2	10400
Headphones	2	8400
Bluetooth Speaker	1	3200
USB Cable	6	2700
Mechanical Keyboard	1	2500

PRODUCT_NAME	QUANTITY_SOLD	TOTAL_REVENUE
Laptop Stand	1	1500
Wireless Mouse	1	850

13 rows selected.

Report 3: Payment Analysis Report

Display:

- Payment mode
- Number of transactions
- Successful payments

```
SQL> SELECT
  2     PAYMENTMODE AS PAYMENT_MODE,
  3     COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  4     SUM(
  5         CASE
  6             WHEN PAYMENTSTATUS = 'Completed' THEN 1
  7             ELSE 0
  8         END
  9     ) AS SUCCESSFUL_PAYMENTS
10 FROM PAYMENT
11 GROUP BY PAYMENTMODE
12 ORDER BY NUMBER_OF_TRANSACTIONS DESC;
```

PAYMENT_MODE	NUMBER_OF_TRANSACTIONS	SUCCESSFUL_PAYMENTS
Debit Card	3	2
UPI	2	2
Credit Card	2	2
COD	1	1
Net Banking	1	1

Report 4: Customer Purchase Analysis

Find:

- Top purchasing customers

```
SQL> SELECT
  2     C.CUSTOMER_ID,
  3     C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4     SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
  5 FROM CUSTOMER C
  6 JOIN ORDERS O
  7     ON C.CUSTOMER_ID = O.CUSTOMERID
  8 GROUP BY
  9     C.CUSTOMER_ID,
10     C.FIRST_NAME,
11     C.LAST_NAME
12 ORDER BY TOTAL_SPENDING DESC;
```

CUSTOMER_I	CUSTOMER_NAME	TOTAL_SPENDING
CUST000001	Arjun Kumar	55850
CUST000014	Lakshmi Venkatesan	28000
CUST000003	Rahul Menon	28000
CUST000005	Vikram Reddy	22000
CUST000009	Karthik Subramanian	18500
CUST000015	Nithin Murugan	8350
CUST000004	Ananya Iyer	8300
CUST000006	Meera Krishnan	7000
CUST000011	Aditya Verma	7000
CUST000007	Suresh Nair	6500
CUST000008	Kavya Rajan	5700

CUSTOMER_I	CUSTOMER_NAME	TOTAL_SPENDING
CUST000010	Divya Pillai	5200
CUST000002	Priya Sharma	3350
CUST000012	Sneha Patel	2450

14 rows selected.

- Customers with maximum orders

```

SQL> SELECT
2  C.CUSTOMER_ID,
3  C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4  COUNT(O.ORDERID) AS NUMBER_OF_ORDERS
5  FROM CUSTOMER C
6  JOIN ORDERS O
7  ON C.CUSTOMER_ID = O.CUSTOMERID
8  GROUP BY
9  C.CUSTOMER_ID,
10 C.FIRST_NAME,
11 C.LAST_NAME
12 HAVING COUNT(O.ORDERID) = (
13   SELECT MAX(ORDER_COUNT)
14   FROM (
15    SELECT COUNT(*) AS ORDER_COUNT
16    FROM ORDERS
17    GROUP BY CUSTOMERID
18   )
19 );

```

CUSTOMER_ID	CUSTOMER_NAME	NUMBER_OF_ORDERS
CUST000001	Arjun Kumar	1
CUST000002	Priya Sharma	1
CUST000003	Rahul Menon	1
CUST000004	Ananya Iyer	1
CUST000005	Vikram Reddy	1
CUST000006	Meera Krishnan	1
CUST000007	Suresh Nair	1
CUST000008	Kavya Rajan	1
CUST000009	Karthik Subramanian	1
CUST000010	Divya Pillai	1
CUST000011	Aditya Verma	1
CUST000012	Sneha Patel	1
CUST000014	Lakshmi Venkatesan	1
CUST000015	Nithin Murugan	1

14 rows selected.

● Customers with highest spending

```

SQL> SELECT
2  C.CUSTOMER_ID,
3  C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4  SUM(O.TOTALAMOUNT) AS TOTAL_SPENDING
5  FROM CUSTOMER C
6  JOIN ORDERS O
7  ON C.CUSTOMER_ID = O.CUSTOMERID
8  GROUP BY
9  C.CUSTOMER_ID,
10 C.FIRST_NAME,
11 C.LAST_NAME
12 HAVING SUM(O.TOTALAMOUNT) = (
13   SELECT MAX(TOTAL_SPENDING)
14   FROM (
15    SELECT SUM(TOTALAMOUNT) AS TOTAL_SPENDING
16    FROM ORDERS
17    GROUP BY CUSTOMERID
18   )
19 );

```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_SPENDING
CUST000001	Arjun Kumar	55850